Learning Annotation-Driven Latent Structure for Multi-Modal Fusion

Comments:

This paper proposes MMCTL for cross-modal alignment of heterogeneous remote-sensing data using annotation-driven continuous similarity. The topic is relevant and the results are promising. However, the distinction from existing contrastive and cross-modal metric-learning methods should be clarified, and stronger baselines, generalization tests, parameter analysis, and statistical validation are recommended. Overall, I think the method is too simple and the contributions are limited, and the suggestions are as follows:

1、Please clarify the essential difference between MMCTL and supervised contrastive learning or cross-modal triplet loss, especially the added contribution of continuous similarity constraints.

2、The paper claims that MMCTL learns a more interpretable latent space; visualizations or cross-modal retrieval examples are recommended for direct validation.

3、Please specify the rules for selecting positive and negative samples in continuous triplets, as well as the similarity thresholds or margin settings.

4、How are annotation attributes such as category, object size, and height normalized and weighted? Does the weighting strategy affect the final results?

5、What is the rationale for freezing the encoder in the two-stage training scheme? A comparison with end-to-end fine-tuning would help validate this design.

6、The baseline set is limited. Representative multimodal methods, such as supervised contrastive learning, cross-attention, or Transformer-based fusion, should be included.

7、Please report standard deviations and statistical significance tests to verify whether MMCTL's improvements over the baselines are stable and reliable.